Bayesian Networks: Representation

CS5491: Artificial Intelligence

ZHICHAO LU

Content Credits: Prof. Wei's CS4486 Course and Prof. Boddeti's AI Course

TODAY

Independence

Bayes Net Introduction

PROBABILISTIC MODELS

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What do we do with probabilistic models?

- We (or our agents) need to reason about unknown variables, given evidence
- Example: explanation (diagnostic reasoning)
- Example: prediction (causal reasoning)
- Example: value of information

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Independence is a simplifying *modeling assumption*

- *Empirical* joint distributions: at best "close" to independence
- What could we assume for (Weather, Traffic, Cavity, Toothache)?

EXAMPLE: INDEPENDENCE

T	W	P
hot	sun	0.4
hot	rain	0.1
cold	sun	0.3
cold	rain	0.2

EXAMPLE: INDEPENDENCE

T	P
hot	0.5
cold	0.5

W	P
sun	0.6
rain	0.4

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$P(X_n)$

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$\cdots$

$P(X_n)$

H	0.5
T	0.5

$P(X_1, \ldots, X_n)$

X_1	X_2	$\cdots$	X_n	
H	H	$\cdots$	H	
H	H	$\cdots$	T	
$\vdots$	$\vdots$	$\vdots$	$\vdots$	
T	T	$\cdots$	T	

CONDITIONAL INDEPENDENCE

P(Toothache, Cavity, Catch)

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The same independence holds if I do not have a cavity:

- $P(+catch|+toothache, -cavity) = P(+catch|-cavity)$

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Catch is conditionally independent of Toothache given Cavity:

- $P(catch|toothache, cavity) = P(catch|cavity)$

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Equivalent statements:

- $P(toothache|catch, cavity) = P(toothache|cavity)$
- $P(toothache, catch|cavity) = P(toothache|cavity)P(catch|cavity)$
- One can be derived from the other easily

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- iff:

$$\forall x, y, z: \; P(x, y|z) = P(x|z)P(y|z)$$

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- iff:

$$\forall x, y, z: \; P(x, y|z) = P(x|z)P(y|z)$$

- or, equivalently, iff

$$\forall x, y, z: \; P(x|y, z) = P(x|z)$$

CONDITIONAL INDEPENDENCE

Example 1:

- T: Traffic
- U: Umbrella
- R: Raining

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- T: Traffic
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Example 2:

- F: Fire
- S: Smoke
- A: Alarm

CONDITIONAL INDEPENDENCE AND CHAIN RULE

Chain rule: $P(X_1, X_2, \ldots, X_n) = P(X_1)P(X_2|X_1)P(X_3|X_2, X_1)\ldots$

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Bayes nets / graphical models help us express conditional independence assumptions

BAYES NET: BIG PICTURE

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Two problems with using full joint distribution tables as our probabilistic models:

- Unless there are only a few variables, the joint is WAY too big to represent explicitly
- Hard to learn (estimate) anything empirically about more than a few variables at a time

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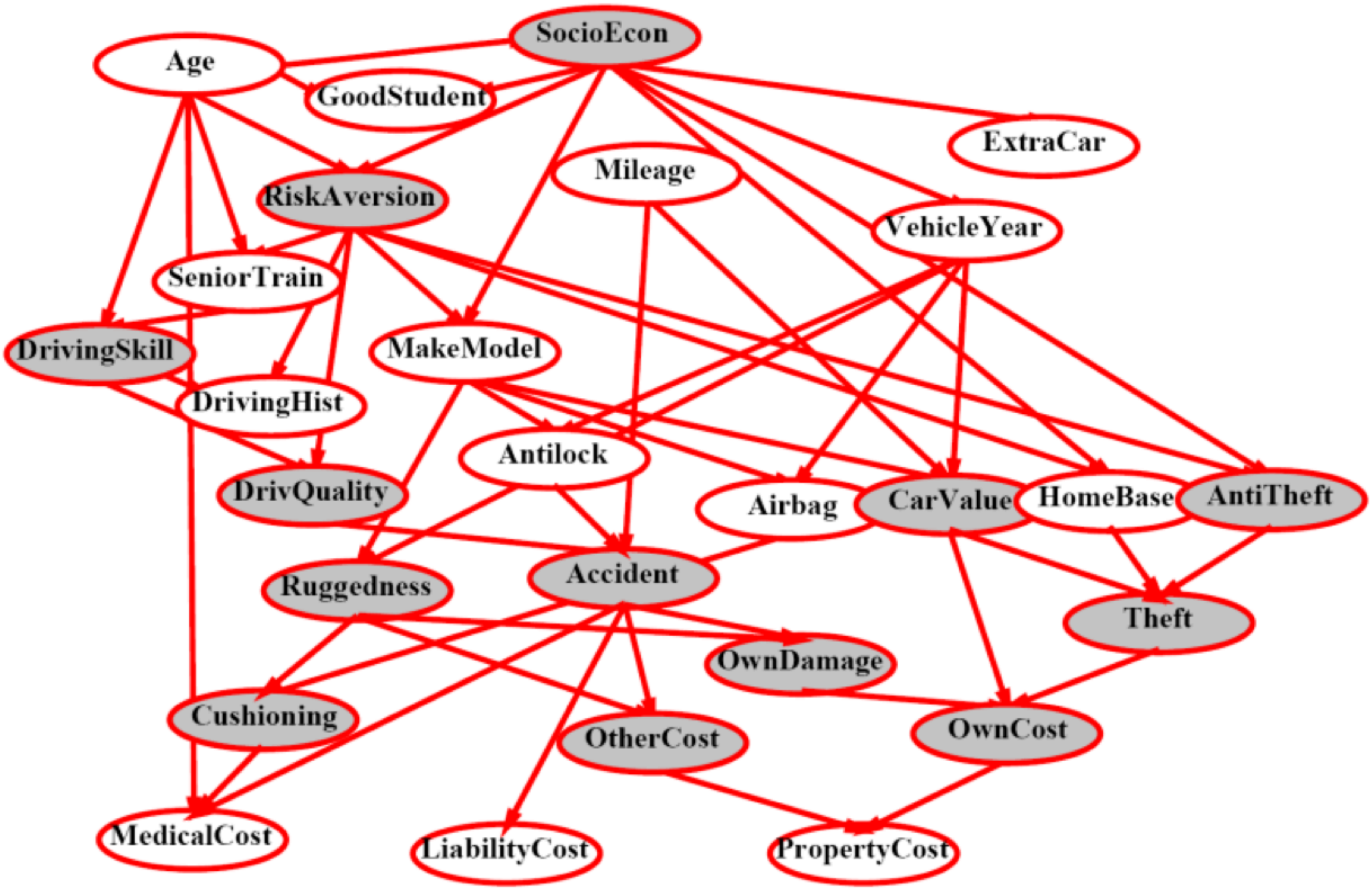
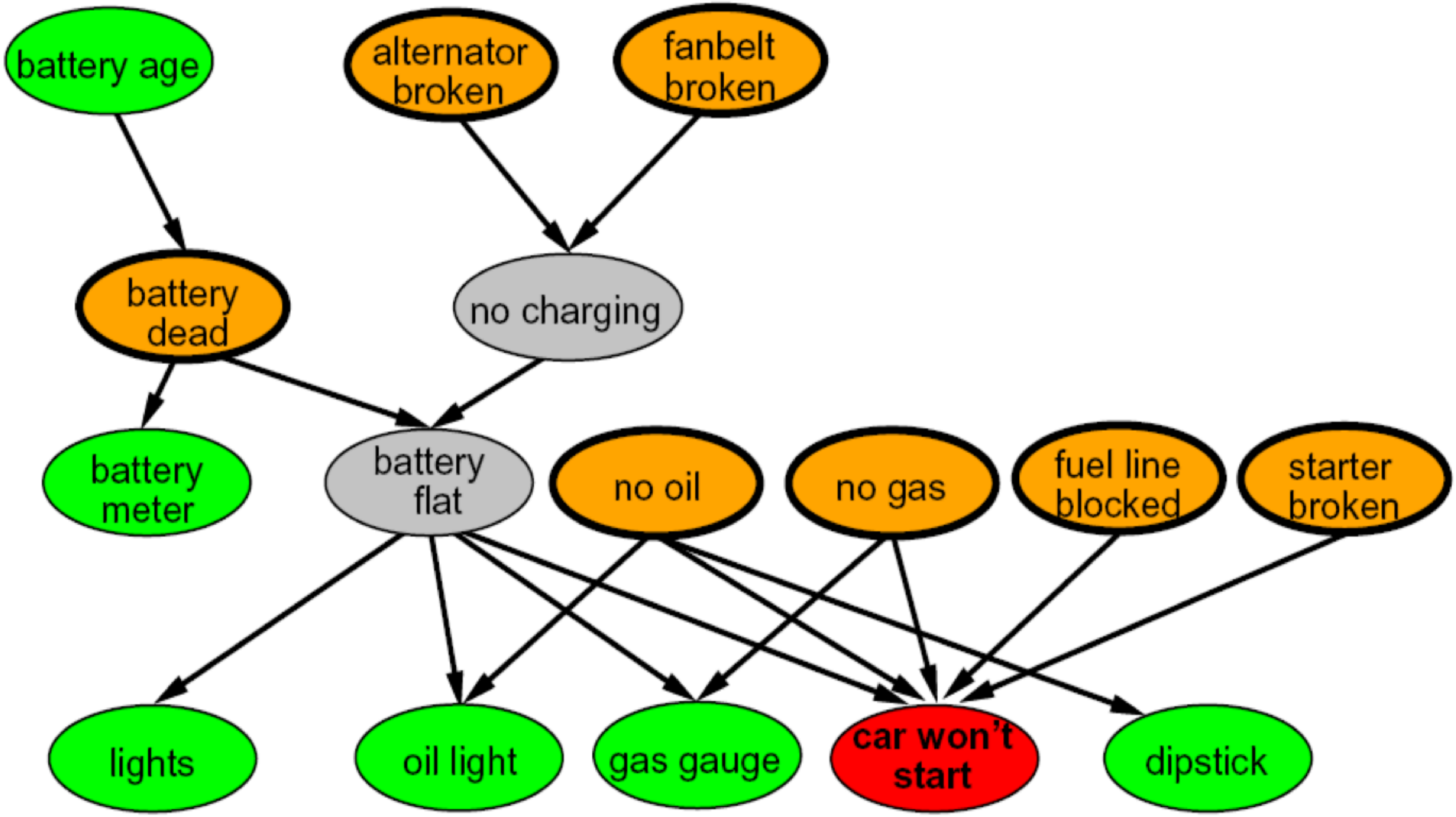
Bayes nets: a technique for describing complex joint distributions (models) using simple, local distributions (conditional probabilities)

- More properly called probabilistic graphical models
- We describe how variables locally interact
- Local interactions chain together to give global, indirect interactions

EXAMPLE BAYES NET: INSURANCE



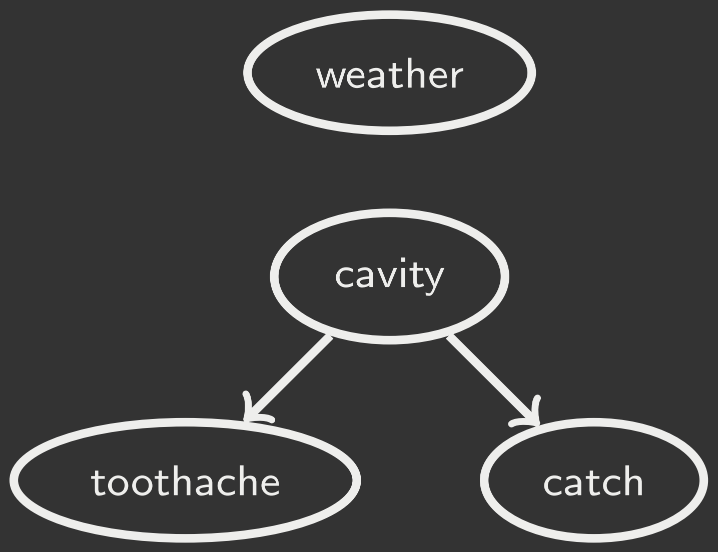
EXAMPLE BAYES NET: CAR



GRAPHICAL MODEL NOTATION

Nodes: variables (with domains)

- Can be assigned (observed) or unassigned (unobserved)



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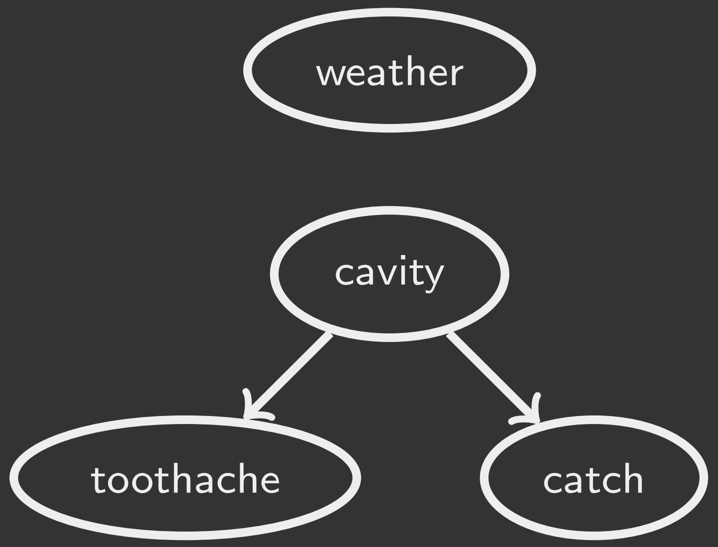
- Similar to CSP constraints
- Indicate "direct influence" between variables
- Formally: encode conditional independence (more later)



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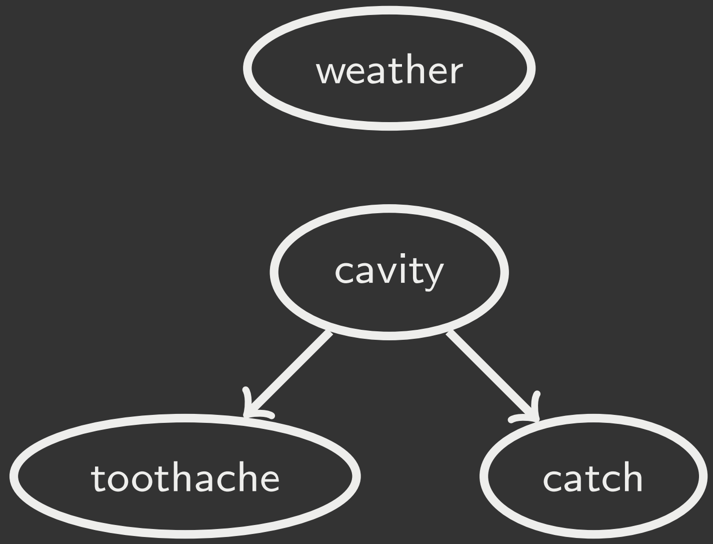
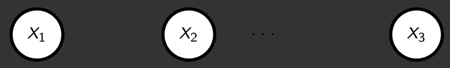


For now: imagine that arrows mean direct causation (in general, they don't.)

EXAMPLE: COIN FLIPS

N independent coin flips



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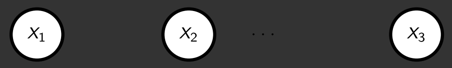

No interactions between variables: absolute independence

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- R: it rains
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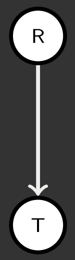
Model 2: rain causes traffic



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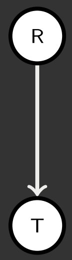

Why is an agent using model 2 better?

EXAMPLE: TRAFFIC II

How do we build a causal graphical model?

Variables:

- T: Traffic
- R: It rains
- L: Low pressure
- D: Roof drips
- B: Ballgame
- C: Cavity

EXAMPLE: ALARM NETWORK

How do we build a causal graphical model?

Variables:

- B: Burglary
- A: Alarm goes off
- M: Mary calls
- J: John calls
- E: Earthquake

BAYES NET SEMANTICS

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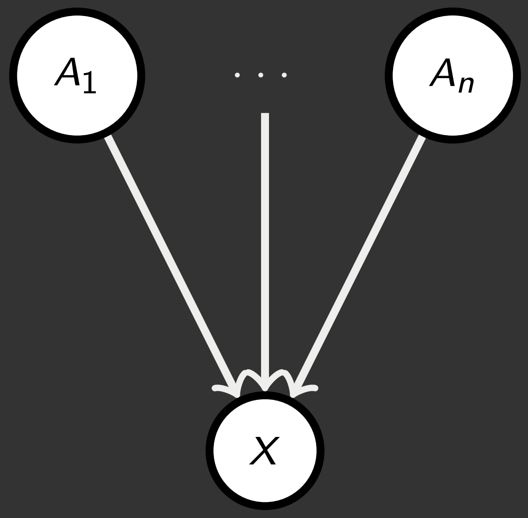
A set of nodes, one per variable X



$P(X|A_1, \ldots, A_n)$

BAYES NET SEMANTICS

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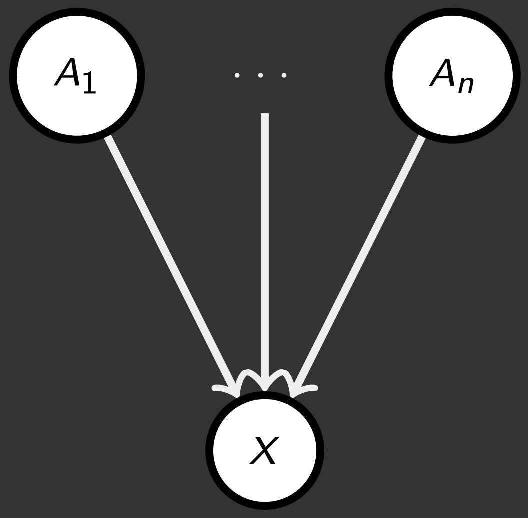

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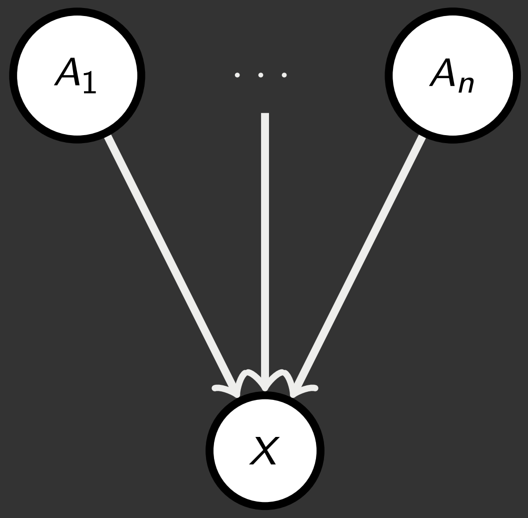

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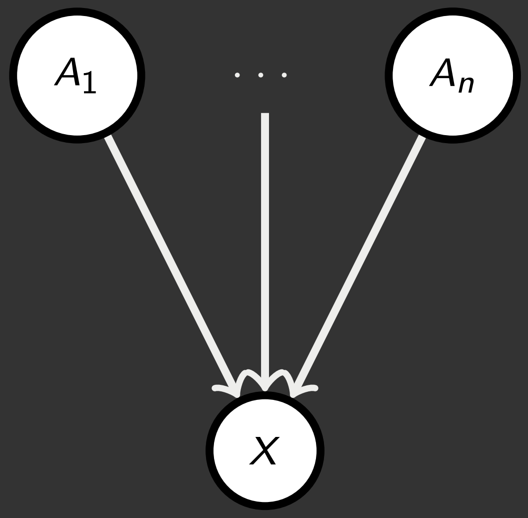

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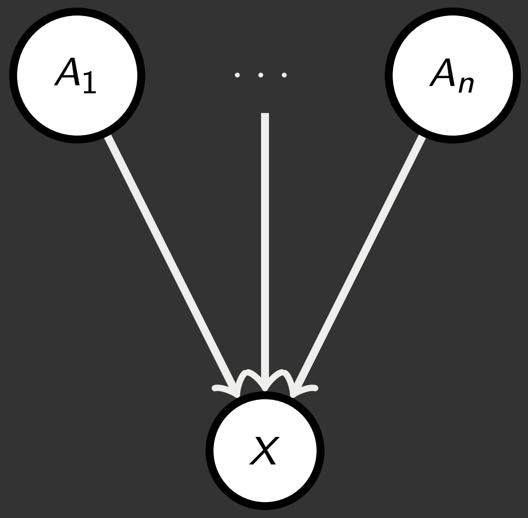

$P(X|A_1, \ldots, A_n)$

A Bayes net = Topology (graph) + Local Conditional Probabilities

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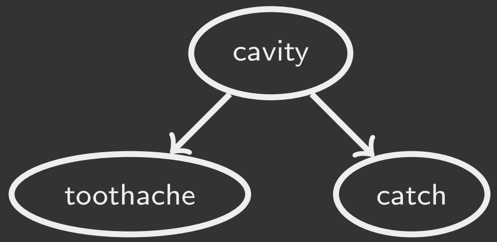
- Example: $P(+cavity, +catch, -toothache)$



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Not every BN can represent every joint distribution

- The topology enforces certain conditional independencies

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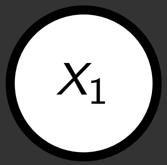

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EXAMPLE: COIN FLIPS

X_1 X_2 $\cdots$ X_n

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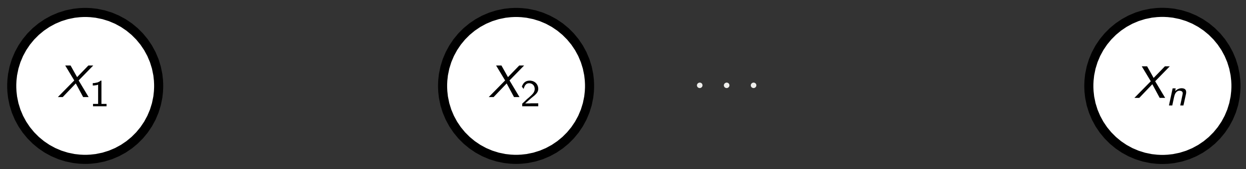

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$$P(h, h, t, h) =$$

Only distributions whose variables are absolutely independent can be represented by a Bayes net with no arcs.

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$P(R)$

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+t	1/2
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$$P(+r,-t) =$$

EXAMPLE: TRAFFIC

Causal direction

$P(T)$

+r	1/4
-r	3/4

$P(R|T=+t)$

+t	3/4
-t	1/4

$P(R|T=-t)$

+t	1/2
-t	1/2

+r	+t	3/16
+r	-t	1/16
-r	+t	6/16
-r	-t	6/16

EXAMPLE: REVERSE TRAFFIC

Reverse Causality?

$P(T)$

+t	9/16
-t	7/16

$P(R|T=+t)$

+r	1/3
-r	2/3

$P(R|T=-t)$

+r	1/7
-r	6/7

+r	+t	3/16
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What do the arrows really mean?

- Topology may happen to encode causal structure
- Topology really encodes conditional independence

$$P(x_i|x_1,\ldots,x_{i-1}) = P(x_i|parents(x_i))$$

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So far: how a Bayes net encodes a joint distribution

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  - First assembled BNs using an intuitive notion of conditional independence as causality
  - Then saw that key property is conditional independence
- Main goal: answer queries about conditional independence and influence

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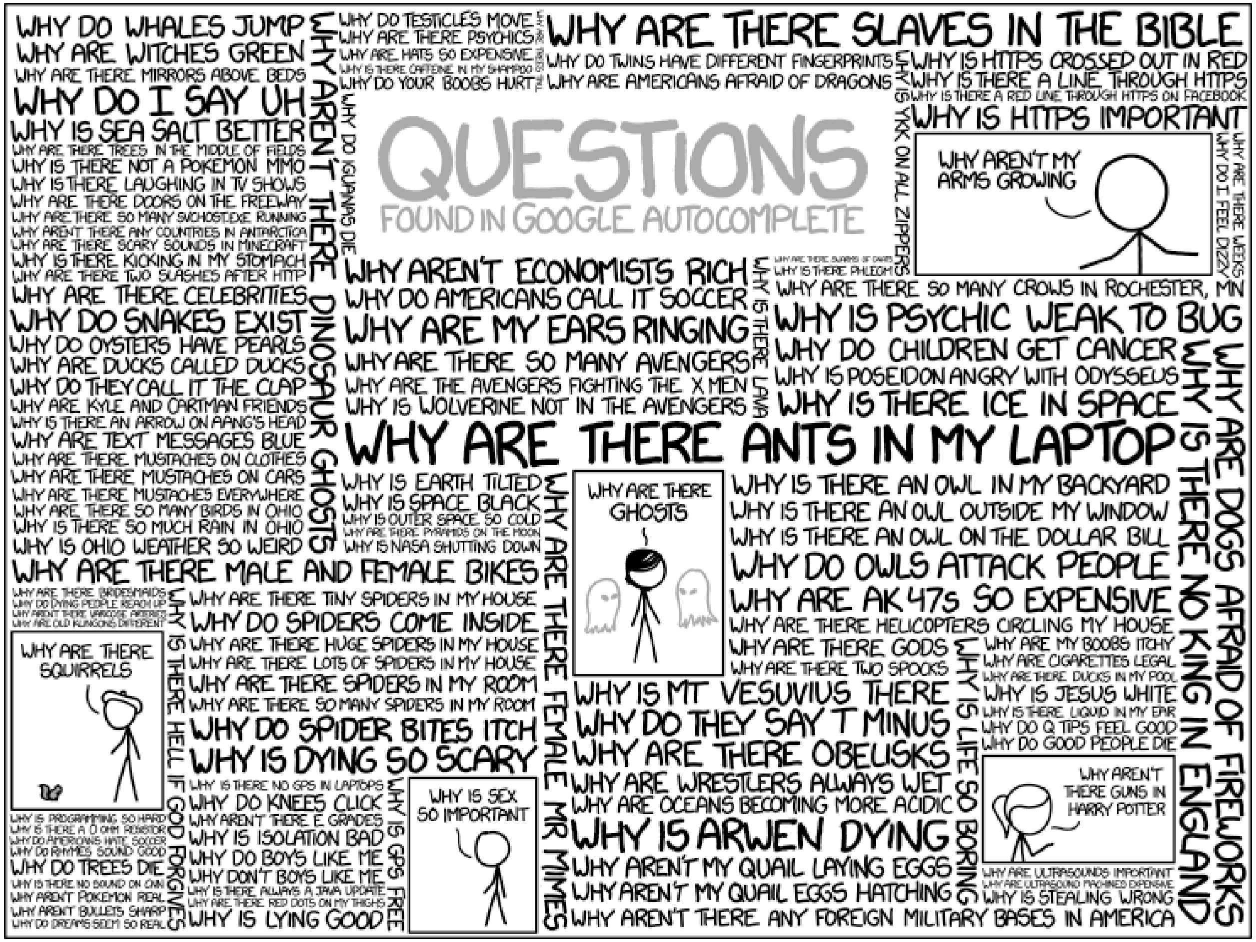
After that: how to answer numerical queries (inference)

Q & A



WHY DO WHALES JUMP
WHY ARE WITCHES GREEN
WHY ARE THERE MIRRORS ABOVE BEDS
WHY DO I SAY UH
WHY IS SEA SALT BETTER
WHY ARE THERE TREES IN THE MIDDLE OF FIELDS
WHY IS THERE NOT A POKEMON MMO
WHY IS THERE LAUGHING IN TV SHOWS
WHY ARE THERE DOORS ON THE FREEWAY
WHY ARE THERE SO MANY SVCHOST.EXE RUNNING
WHY AREN'T THERE ANY COUNTRIES IN ANTARCTICA
WHY ARE THERE SCARY SOUNDS IN MINECRAFT
WHY IS THERE KICKING IN MY STOMACH
WHY ARE THERE TWO SLASHES AFTER HTTP
WHY ARE THERE CELEBRITIES
WHY DO SNAKES EXIST
WHY DO OYSTERS HAVE PEARLS
WHY ARE DUCKS CALLED DUCKS
WHY DO THEY CALL IT THE CLAP
WHY ARE KYLE AND CARTMAN FRIENDS
WHY IS THERE AN ARROW ON AANG'S HEAD
WHY ARE TEXT MESSAGES BLUE
WHY ARE THERE MUSTACHES ON CLOTHES
WHY ARE THERE MUSTACHES ON CARS
WHY ARE THERE MUSTACHES EVERYWHERE
WHY ARE THERE SO MANY BIRDS IN OHIO
WHY IS THERE SO MUCH RAIN IN OHIO
WHY IS OHIO WEATHER SO WEIRD
WHY AREN'T THERE DINOSAUR GHOSTS
WHY DO IGUANAS DIE
WHY DO TESTICLES MOVE
WHY ARE THERE PSYCHICS
WHY ARE HATS SO EXPENSIVE
WHY IS THERE CAFFEINE IN MY SHAMPOO
WHY DO YOUR BOOBS HURT
WHY ARE THERE SLAVES IN THE BIBLE
WHY DO TWINS HAVE DIFFERENT FINGERPRINTS
WHY ARE AMERICANS AFRAID OF DRAGONS
WHY IS HTTPS CROSSED OUT IN RED
WHY IS THERE A LINE THROUGH HTTPS
WHY IS THERE A RED LINE THROUGH HTTPS ON FACEBOOK
WHY IS HTTPS IMPORTANT
WHY IS YKK ON ALL ZIPPERS
QUESTIONS
FOUND IN GOOGLE AUTOCOMPLETE
WHY AREN'T MY ARMS GROWING
WHY ARE THERE WEEKS
WHY DO I FEEL DIZZY
WHY AREN'T ECONOMISTS RICH
WHY DO AMERICANS CALL IT SOCCER
WHY ARE MY EARS RINGING
WHY ARE THERE SO MANY AVENGERS
WHY ARE THE AVENGERS FIGHTING THE X MEN
WHY IS WOLVERINE NOT IN THE AVENGERS
WHY IS THERE LAVA
WHY IS THERE PHLEGM
WHY ARE THERE SO MANY CROWS IN ROCHESTER, MN
WHY IS PSYCHIC WEAK TO BUG
WHY DO CHILDREN GET CANCER
WHY IS POSEIDON ANGRY WITH ODYSSEUS
WHY IS THERE ICE IN SPACE
WHY ARE THERE ANTS IN MY LAPTOP
WHY IS EARTH TILTED
WHY IS SPACE BLACK
WHY IS OUTER SPACE SO COLD
WHY ARE THERE PYRAMIDS ON THE MOON
WHY IS NASA SHUTTING DOWN
WHY ARE THERE GHOSTS
WHY IS THERE AN OWL IN MY BACKYARD
WHY IS THERE AN OWL OUTSIDE MY WINDOW
WHY IS THERE AN OWL ON THE DOLLAR BILL
WHY DO OWLS ATTACK PEOPLE
WHY ARE AK 47s SO EXPENSIVE
WHY ARE THERE HELICOPTERS CIRCLING MY HOUSE
WHY ARE THERE GODS
WHY ARE THERE TWO SPOCKS
WHY ARE DOGS AFRAID OF FIREWORKS
WHY IS THERE NO KING IN ENGLAND
WHY ARE THERE MALE AND FEMALE BIKES
WHY ARE THERE BRIDESMAIDS
WHY DO DYING PEOPLE REACH UP
WHY AREN'T THERE VARICOSE ARTERIES
WHY ARE OLD KLINGONS DIFFERENT
WHY ARE THERE SQUIRRELS
WHY IS THERE HELL IF GOD FORGIVES
WHY ARE THERE TINY SPIDERS IN MY HOUSE
WHY DO SPIDERS COME INSIDE
WHY ARE THERE HUGE SPIDERS IN MY HOUSE
WHY ARE THERE LOTS OF SPIDERS IN MY HOUSE
WHY ARE THERE SPIDERS IN MY ROOM
WHY ARE THERE SO MANY SPIDERS IN MY ROOM
WHY DO SPIDER BITES ITCH
WHY IS DYING SO SCARY
WHY ARE THERE FEMALE MR MIMES
WHY IS MT VESUVIUS THERE
WHY DO THEY SAY T MINUS
WHY ARE THERE OBELISKS
WHY ARE WRESTLERS ALWAYS WET
WHY ARE OCEANS BECOMING MORE ACIDIC
WHY IS ARWEN DYING
WHY AREN'T MY QUAIL LAYING EGGS
WHY AREN'T MY QUAIL EGGS HATCHING
WHY AREN'T THERE ANY FOREIGN MILITARY BASES IN AMERICA
WHY IS LIFE SO BORING
WHY ARE MY BOOBS ITCHY
WHY ARE CIGARETTES LEGAL
WHY ARE THERE DUCKS IN MY POOL
WHY IS JESUS WHITE
WHY IS THERE LIQUID IN MY EAR
WHY DO Q TIPS FEEL GOOD
WHY DO GOOD PEOPLE DIE
WHY AREN'T THERE GUNS IN HARRY POTTER
WHY ARE ULTRASOUNDS IMPORTANT
WHY ARE ULTRASOUND MACHINES EXPENSIVE
WHY IS STEALING WRONG
WHY IS PROGRAMMING SO HARD
WHY IS THERE A 0 OHM RESISTOR
WHY DO AMERICANS HATE SOCCER
WHY DO RHYMES SOUND GOOD
WHY DO TREES DIE
WHY IS THERE NO SOUND ON CNN
WHY AREN'T POKEMON REAL
WHY AREN'T BULLETS SHARP
WHY DO DREAMS SEEM SO REAL
WHY IS THERE NO GPS IN LAPTOPS
WHY DO KNEES CLICK
WHY AREN'T THERE E GRADES
WHY IS ISOLATION BAD
WHY DO BOYS LIKE ME
WHY DON'T BOYS LIKE ME
WHY IS THERE ALWAYS A JAVA UPDATE
WHY ARE THERE RED DOTS ON MY THIGHS
WHY IS LYING GOOD
WHY IS GPS FREE
WHY IS SEX SO IMPORTANT

XKCD